

Testing the significance of the entire model:

- is possible only in non-parametric models
- allows to assess whether the estimated model is able to explain the variability of the studied phenomenon
- consists in testing the hypothesis that all parameters of the model (including the constant) are equal to zero at the same time
- allows to assess which of the variables have a significant impact on the studied phenomenon

Which of the following statements regarding the information criteria is NOT true:

- information criteria impose an additional "penalty" on the optimization criterion for the size of the model
- information criteria can be used to select the best model
- the lower the value of the information criterion, the worse the model
- the use of the BIC criterion often results in a model with fewer variables than in case of AIC

The problem of overfitting:

- may appear for too flexible model
- means a monotonic increase in the prediction error on the training sample
- means a monotonic increase in the forecast error on the test sample
- means estimating the model on too many samples

Which of the following statements regarding cross-validation is NOT true:

- cross-validation means randomly dividing the training sample into smaller samples and estimating the analyzed model several times
- cross-validation can be used to tune model hyperparameters (search for their optimal values)
- cross-validation can be used to estimate the prediction error on the new data
- cross-validation means randomly dividing the test sample into smaller samples and estimating the analyzed model several times

The LASSO method:

- is an algorithm used exclusively to solve regression problems
- can be used as a method of selecting variables for the model
- requires rescaling the variables to the range 0-1
- consists in dividing the sample into a larger number of subgroups in order to obtain a more accurate match

Which of the following statements regarding leave one out validation (LOOCV) is NOT true:

- LOOCV validation does not require the use of a random number generator
- LOOCV validation is more often used in small datasets
- LOOCV validation may give a different assessment of the forecast error, depending on the division of the learning sample into smaller subsamples used in it
- in LOOCV validation, the average prediction error is calculated as the average of n errors, where n is the size of the training sample